CUMEDI-3030316 Gastrointestinal System Module 2026

Medication in upper GI diseases [In-class materials]

Case 1

Mr. Somchai is a 52 years old advertising executive who presents to the emergency room complaining of the passage of black stools for 3 days and associated lightheadedness. He also relates that he cannot keep up with his usual schedule because of fatigue.

Upon further questioning he states that his stools are not only black, but are sticky and malodorous. He further complains of recent worsening of a chronic epigastric burning which had been a problem off/on for 8 months. Mr. Somchai has been taking NSAIDS as needed for back pain and knee pain during daily activities for years.

Mr. Somchai has recently taken more NSAIDS as needed for a Colles' fracture that happened last week. Aside from that, his weight is stable and he claims to have an excellent appetite. He has a normal bowel habit and has not had prior black stools.

Initial physical examination reveals an alert, oriented, overweight male. He appears pale, anxious and somewhat restless. Vital signs are as follows: Blood Pressure 100/70 mmHg, Heart Rate 110/min, Respiratory Rate 20/minute, Temperature 37 C. The abdominal examination reveals a rounded abdomen. There is moderate tenderness in the epigastrium. The exam was felt to be suboptimal secondary to the patient's obesity. Rectal examination revealed black, tarry stool. Complete blood count reveals the following: Hemoglobin 7.8 g/dL, Hematocrit 25%, MCV 68 fL, MCH 21 pg, MCHC 31 g/dL, RDW 18.5%, WBC 8,900/µL, and platelet count 420,000/µL.

Esophagogastroduodenoscopy was performed. Two 1 cm gastric ulcer with black pigmented spot based was seen. Rapid urease test was performed and found to be positive.

Mr. Somchai was given medication for his condition.

QUESTIONS:

1. **Explain the association of NSAID and gastric ulcer.**

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2. If Mr. Somchai needs to use NSAIDs again after this event, what kinds of NSAIDs would you recommend and why?

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3. What is the meaning of positive Rapid urease test and what is the mechanism of action of medication that you would use for this patient?

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4. What are the medications should Mr. Somchai took and the side effects to be aware of from his medication?

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Case 2

Mr. Ben, a 34-year-old man presents to his primary care provider with a burning in his upper and middle chest, as well as the development of a sour taste in his mouth over the last several months. He further describes his symptoms as occurring 30 minutes after eating and lasting up to 90 minutes, as well as occurring at night.

Ben has had no prior surgeries or other significant health concerns and no family history of malignancy. He states he does not use tobacco or drugs and that he only consumes 1 to 2 glasses of wine on the weekend. He has not experienced weight loss and is not aware of food hypersensitivities.

Ben was diagnosed with gastroesophageal reflux disease (GERD). Acid lowering agents were prescribed.

QUESTIONS:

1. List the classes of drugs that you would use, and explain the mechanism of action for each type related to physiology class you have learnt

2. State the adverse effects associated with each class?

Case 3

Mr. Vichai, a 52-year-old man with a history of active alcoholism presented to the emergency room with hematemesis and passage of bright red stools that began 8 hours before. He claimed that after drinking at a friend's party, he began vomiting blood several times and started to feel dizzy.

On initial examination, the patient was drowsy. He had a pale tone, tachycardic (140 bpm), hypotensive (95/56 mm Hg) but still breathing spontaneously (RR 20/min). Apart from icteric sclerae, few spider nevi on chest wall and a palpable spleen, other physical examination was unremarkable. Per rectal examination confirms bright red bleed.

An EGD revealed three large esophageal varices, each with a red whale and a white nipple sign, indicating recent esophageal varices bleeding. After the bleeding was stopped with esophageal banding, the doctor prescribed continuous intravenous synthetic hormonal analog for three days. Vichai was released from the hospital after five days with an oral sympatholytic drug prescription.

QUESTIONS:

1. What is the intravenous synthetic hormonal analog that this patient got?

2. Explain how the medication helps with variceal bleeding.

3. What are the potential side effects of Octreotide?

4. Describe how an oral sympatholytic drug works to prevent variceal bleeding. And what is it that was prescribed to the patient?

5. What should Vichai be aware of while using a sympatholytic drug?